

# WASTEWATER TREATMENT USING TRANSFORMER MPC MODEL

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## ABSTRACT

Wastewater treatment plants (WWTPs) are highly nonlinear, dynamic systems governed by complex biological processes, making conventional control strategies insufficient for maintaining consistent effluent quality while minimizing energy consumption. This paper proposes a Transformer-based Model Predictive Control (MPC) framework for intelligent optimization of WWTP operation, with a focus on aeration control and pollutant regulation. A deep-learning Transformer model is trained on historical WWTP time-series data to capture long-term temporal dependencies and perform multi-step prediction of key parameters such as ammonium ( $\text{NH}_4$ ), nitrate ( $\text{NO}_3$ ), dissolved oxygen ( $\text{O}_2$ ), phosphate ( $\text{PO}_4$ ), suspended solids (SS), and derived chemical and biological oxygen demands (COD and BOD). These predictions are embedded within an MPC controller that computes optimal dissolved oxygen setpoints over a receding horizon, minimizing a cost function that balances pollutant reduction, oxygen usage, and system stability. The closed-loop control scheme is implemented and evaluated in simulation, demonstrating stable  $\text{NH}_4$  and SS behavior, adaptive  $\text{O}_2$  regulation, and improved effluent quality compared to conventional strategies. The proposed Transformer-MPC framework is suitable for deployment in MATLAB/Simulink environments such as the BSM1 benchmark, offering a data-driven, energy-efficient solution for smart wastewater treatment plants.

## INTRODUCTION

Wastewater treatment plants (WWTPs) play a critical role in protecting public

health and the environment by removing pollutants from municipal and industrial effluents before discharge into natural water bodies [1]. Rapid urbanization, population growth, and climate variability have increased both the volume and complexity of wastewater, raising the need for intelligent, energy-efficient, and robust treatment processes [1][2]. Conventional treatment strategies often rely on fixed-rule operations and classical feedback controllers, which struggle to handle the high nonlinearity, time-varying influent loads, and multiple conflicting objectives (energy vs. effluent quality) present in modern WWTPs [2][3].

To address these challenges, the first generation of advanced control methods introduced model-based predictive control (MPC) for activated-sludge-based WWTPs. In the early 2000s, researchers proposed linear MPC formulations for DO and nitrate control using the Benchmark Simulation Model No. 1 (BSM1), exploiting its standardized structure for fair comparison of control strategies [2][3]. While these linear MPC schemes could explicitly handle constraints and multi-variable interactions, their main drawback was limited accuracy under large load variations and strong process nonlinearities, since the plant dynamics were approximated with linear models [2][4].

To overcome this limitation, later works introduced nonlinear MPC (NMPC) and economic MPC (EMPC) variants that directly used the full BSM1 nonlinear equations or reduced-order models [4][5]. These approaches improved constraint handling and allowed explicit optimization of energy consumption and effluent quality,

but they came at a significant computational cost, making real-time deployment on industrial control platforms challenging [4][6].

To reduce complexity, researchers then developed hierarchical and collaborative MPC structures for BSM1, where high-level economic objectives were separated from low-level regulatory control loops [5][6]. These hierarchical schemes improved computational tractability and allowed coordination between the aeration basin, clarifier, and sludge line; however, they still required manual selection of tuning parameters and operating modes, limiting their adaptability to unseen disturbances [6][7].

To make MPC more robust and automatic, subsequent studies explored fuzzy-logic and hybrid-control schemes integrated with MPC, using fuzzy rules to switch between different MPC formulations or to adjust tuning parameters online [7][8]. Although these fuzzy-MPC approaches improved robustness to moderate disturbances, they lacked a systematic way to learn complex, high-dimensional dynamic patterns from large datasets, and their rule-bases remained largely heuristic [8][9].

With the rise of data-driven modeling, a new line of work introduced deep-learning-based surrogate models to approximate the BSM1 dynamics or to replace parts of the full-order model for faster prediction [9][10]. These deep-learning models reduced computational load for MPC prediction and enabled end-to-end training on simulation data, but they employed relatively simple architectures such as multilayer perceptrons or shallow recurrent networks, which struggled with long-range temporal dependencies and multi-scale feature extraction in WWTP data [10][11].

To capture longer-term dependencies, researchers began exploring recurrent architectures (LSTM, GRU) and attention-based neural networks for WWTP effluent and state prediction [11][12]. These models improved multi-step forecasting accuracy for key variables such as COD, BOD, DO, and nitrate, and some works embedded them inside predictive control-like frameworks for soft-sensing or set-point optimization [11][12]. However, these LSTM-based models often suffered from high parameter counts, vanishing-gradient issues, and difficulty in parallelizing training, which limited their scalability and real-time performance [12][13].

To address these limitations, the next wave of research adopted Transformer-based deep-learning architectures for wastewater quality monitoring and prediction [13][14]. These multi-stage frameworks, including Time Series Transformers and Temporal Fusion Transformers, demonstrated superior capability in capturing long-range temporal dependencies and multivariate interactions across multiple sensors and process units, enabling accurate multi-step prediction and anomaly detection [13][14]. Despite their modeling power, these Transformer-based methods were initially used only as standalone prediction or monitoring tools, disconnected from closed-loop control, and therefore did not directly influence actuators such as aeration valves or sludge-recycle pumps [14][15].

To bridge this gap, some researchers proposed deep-reinforcement-learning (DRL)-based agents that learn control policies from interaction with BSM1-type simulators, using deep networks to approximate value or policy functions [15][16]. DRL improved the ability to adapt to changing operating conditions and could implicitly optimize long-term objectives

such as energy savings and effluent compliance; however, these methods often required massive training episodes, exhibited limited interpretability, and faced challenges in guaranteeing constraint satisfaction and stability in real-world plants [16][17].

Parallel to these control-oriented advances, tool-oriented works focused on MATLAB/Simulink-based simulation environments for BSM1, developing graphical interfaces and Simulink compatible models that simplified experimentation with various controllers [17][18]. These environments allowed fast prototyping and educational use of BSM1 control strategies, but they mainly supported classical PID or linear MPC schemes, with only limited integration of complex deep learning or Transformer-based models [18][19].

Most recently, a small number of studies reported deep-learning-informed predictive control structures that used neural networks inside MPC prediction blocks, for example by training neural-state-space models or neural MPC surrogates on BSM1 or similar wastewater processes [19][20]. These approaches combined the data-driven flexibility of deep networks with the constraint-handling and receding-horizon optimization of MPC, improving performance under nonlinear and uncertain conditions [19][20]. Nevertheless, existing works have not yet fully integrated Transformer-based deep-learning models into an MPC controller deployed on the full BSM1 benchmark within a MATLAB/Simulink environment, leaving a clear gap in unified, data-driven, constraint-aware control for WWTPs.

Motivated by these developments and limitations, this paper proposes a Transformer-enhanced MPC framework for a wastewater treatment plant based on the

BSM1 benchmark, implemented and validated in MATLAB/Simulink. The Transformer model is used to predict future states and effluent quality, while the MPC layer optimizes manipulated variables such as aeration rates and sludge-recycle flows under operational constraints. By tightly coupling a high-capacity, sequence-aware Transformer predictor with an MPC optimization engine, the proposed approach aims to simultaneously improve prediction accuracy, energy efficiency, and effluent-quality compliance under realistic influent disturbances, while remaining deployable within an industrial grade simulation environment.

**KEYWORDS:** *Wastewater treatment plant, Transformer model, deep learning, Model Predictive Control (MPC), receding horizon control, aeration control, dissolved oxygen, biological nutrient removal, BSM1, MATLAB/Simulink.*

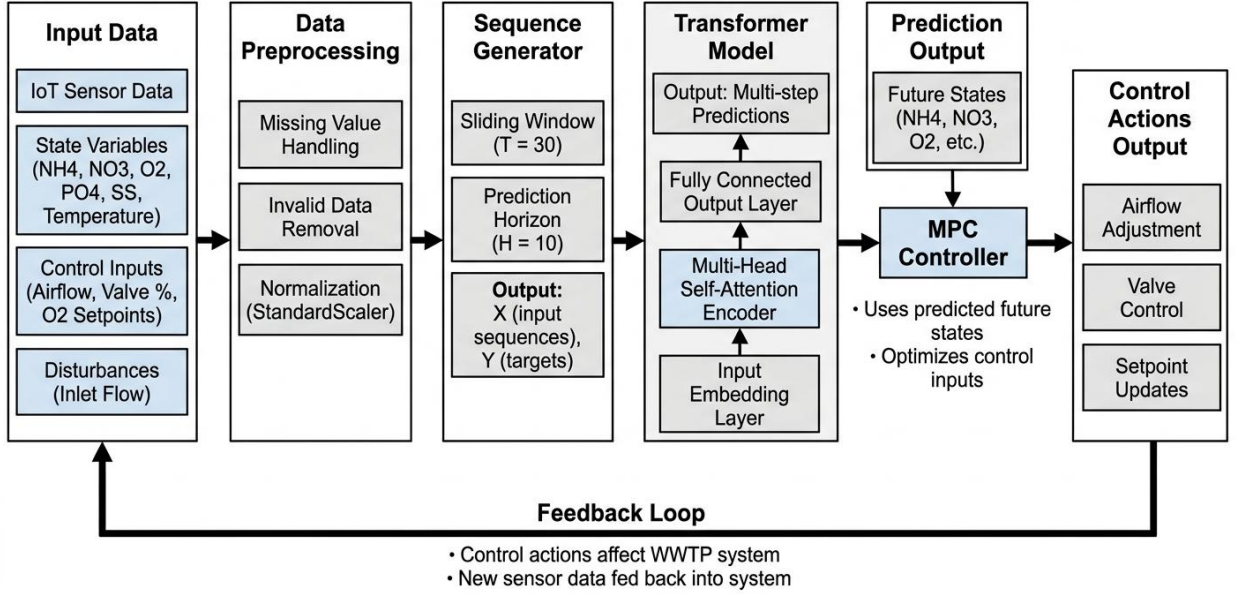
## **PROBLEM STATEMENT**

Wastewater treatment plants must maintain concentrations of key pollutants—such as ammonium ( $\text{NH}_4$ ), nitrate ( $\text{NO}_3$ ), phosphate ( $\text{PO}_4$ ), suspended solids (SS), chemical oxygen demand (COD), and biological oxygen demand (BOD)—within strict environmental limits. However, the underlying biological processes are highly nonlinear, time-varying, and sensitive to fluctuating influent conditions, microbial activity, and operational disturbances, which makes accurate modeling and control challenging. Traditional control methods, including fixed-setpoint or rule-based strategies, often fail to capture these dynamics, leading to excessive aeration energy consumption, unstable pollutant levels, and suboptimal effluent quality. Moreover, most conventional controllers act reactively, adjusting inputs only after deviations occur, rather than anticipating

future process behavior. This motivates the need for an intelligent predictive control framework that can accurately forecast WWTP dynamics and use these predictions to optimize aeration and other control actions in real time. The work addresses this gap by proposing a Transformer-based

MPC control architecture for WWTPs, which combines deep-learning-driven multi-step prediction with closed-loop optimization of dissolved oxygen to achieve stable, energy-efficient, and robust wastewater treatment.

### Transformer-Based Prediction + MPC Framework for WWTP



## V. PROPOSED TRANSFORMER-BASED PREDICTION AND CONTROL FRAMEWORK

In this section, a data-driven prediction and control framework based on a Transformer neural network is developed for the multivariable wastewater treatment plant (WWTP). The proposed method aims to accurately model the nonlinear dynamics of the system and provide reliable multi-step predictions for model predictive control (MPC).

### A. Problem Formulation

The WWTP can be described as a discrete-time nonlinear system as follows:

$$s_{k+1} = F(s_k, u_k, d_k), k = 0, 1, 2, \dots \quad (1)$$

where  $s_k \in \mathbb{R}^n$ ,  $u_k \in \mathbb{R}^m$ , and  $d_k \in \mathbb{R}^p$  denote the system state, control input, and disturbance vector at time step  $k$ , respectively. The nonlinear function  $F(\cdot)$  is unknown and difficult to model explicitly due to the complex biochemical processes and external disturbances in WWTPs.

The control objective is to ensure that the system states track the desired reference trajectory  $r_k$ , which can be expressed as

$$r_k = H(k) \quad (2)$$

where  $H(\cdot)$  denotes the predefined set-point function. The tracking error is defined as

$$z_k = s_k - r_k. \quad (3)$$

## B. Data Preprocessing and Variable Definition

To construct a reliable data-driven model, the raw sensor data are preprocessed as follows.

First, invalid placeholder values are removed and missing values are handled using forward filling. Then, the dataset is divided into three categories:

1. **State variables**  $s_k$ : including concentrations of NH<sub>4</sub>, NO<sub>3</sub>, O<sub>2</sub>, PO<sub>4</sub>, suspended solids (SS), and temperature across multiple tanks.

where  $\mu$  and  $\sigma$  denote the mean and standard deviation of the dataset, respectively.

## C. Sequence Construction for Temporal Learning

To capture the temporal dependencies of the WWTP dynamics, a sliding window approach is adopted. For each sample, an input-output pair is constructed as

$$X_k = [\tilde{Z}_{k-T+1}, \dots, \tilde{Z}_k], \quad (6)$$

$$Y_k = [s_{k+1}, s_{k+2}, \dots, s_{k+H}], \quad (7)$$

where  $T$  is the input window length and  $H$  is the prediction horizon. This formulation enables the model to learn multi-step system evolution.

## D. Transformer-Based Prediction Model

To approximate the nonlinear mapping between past observations and future states, a Transformer encoder network is employed. The model consists of an embedding layer, multiple self-attention encoder layers, and a fully connected output layer.

2. **Control variables**  $u_k$ : including airflow rates, valve positions, and dissolved oxygen setpoints.
3. **Disturbance variables**  $d_k$ : including inlet flow measurements.

The combined feature vector is defined as

$$Z_k = [s_k^T, u_k^T, d_k^T]^T. \quad (4)$$

To improve training stability, the data are normalized using standardization:

$$\tilde{Z}_k = \frac{Z_k - \mu}{\sigma} \quad (5)$$

### Algorithm 1: WWTP Transformer Model Construction & Training

#### Require:

Dataset  $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^N$ , where  $X_i \in \mathbb{R}^{T \times F}$ ,  $Y_i \in \mathbb{R}^{H \times S}$ ,  $S = 9$

Hyperparameters:  $d = 64$ ,  $\text{nhead} = 4$ ,  $L = 3$ , epochs  $E = 50$ , batch size  $B = 64$ ,  $lr = 10^{-3}$

#### Phase 1: Data Preparation

- 1: Split  $\mathcal{D}$  into training set  $\mathcal{D}_{\text{train}}$  (50,000 samples) and test set  $\mathcal{D}_{\text{test}}$  (10,000 samples)
- 2: Convert arrays to float32 tensors; construct *DataLoader* with shuffle,  $B = 64$ , pin memory

#### Phase 2: Model Initialisation

- 3: Initialise input projection:  $\phi: \mathbb{R}^F \rightarrow \mathbb{R}^d$  (single Linear layer)
- 4: Stack  $L$  TransformerEncoder layers with  $d_{\text{model}} = d$ ,  $\text{nhead} = 4$
- 5: Define output head  $f_{\text{out}}: \mathbb{R}^d \rightarrow \mathbb{R}^{H \times S}$  (Linear); reshape to  $(B, H, S)$
- 6: Initialise Adam optimiser; loss criterion = MSE

#### Phase 3: Training Loop

- 7: **for**  $e = 1$  **to**  $E$  **do**
- 8:   **for** each mini-batch  $(X_b, Y_b) \in \mathcal{D}_{\text{train}}$  **do**
- 9:     Project input:  $E_b = \phi(X_b)$
- 10:    Encode:  $Z_b = \text{TransformerEncoder}(E_b)$ ; extract last token  $z = Z_b[:, -1, :]$
- 11:    Predict:  $\hat{Y}_b = \text{reshape}(f_{\text{out}}(z), [B, H, S])$
- 12:    Compute loss:  $\mathcal{L} = \text{MSE}(\hat{Y}_b, Y_b)$
- 13:    Back-propagate  $\mathcal{L}$ ; update parameters via Adam
- 14:   **end for**
- 15: **end for**
- 16: Save trained model weights  $\theta^*$  to disk
- 17: **return** trained model  $f_{\theta^*}$

The embedding operation is given by

$$E_k = W_e X_k + b_e \quad (8)$$

where  $W_e$  and  $b_e$  are learnable parameters.

The encoded representation is obtained through a stack of self-attention layers:

$$H_k = \text{Encoder}(E_k) \quad (9)$$

Finally, the predicted future states are computed as

$$\hat{Y}_k = W_o H_k + b_o \quad (10)$$

where  $\hat{Y}_k \in \mathbb{R}^{H \times n}$ .

The Transformer model effectively captures long-range temporal dependencies and nonlinear interactions among variables, making it suitable for WWTP dynamics modeling.

### E. Model Training

The model parameters are optimized by minimizing the mean squared error (MSE) loss function:

$$\mathcal{L} = \frac{1}{N} \sum_{k=1}^N \|Y_k - \hat{Y}_k\|^2 \quad (11)$$

where  $N$  denotes the number of training samples.

The model is trained using the Adam optimizer with mini-batch gradient descent. A subset of the dataset is used for training, while unseen data are reserved for testing to evaluate generalization performance.

### F. Integration with Model Predictive Control

The trained Transformer model is incorporated into the MPC framework to predict future system states over a finite horizon. The control objective is formulated as

$$J = \sum_{i=1}^H \|s_{k+i} - r_{k+i}\|^2 + \lambda \|u_{k+i}\|^2 \quad (12)$$

where  $\lambda$  is a weighting factor that balances tracking performance and control effort.

#### Algorithm 3: MPC Cost Evaluation (Single-Step Lookahead)

**Require:**

Control input  $u = (u_o, \dots, u_{H-1})$ , current window  $x_o \in \mathbb{R}^{1 \times T \times F}$ , model  $f_{\theta^*}$

- 1: Set  $x \leftarrow x_o$
- 2: Apply first control:  $x[:, -1, u_{idx}] \leftarrow u_o$
- 3: Predict:  $\hat{y} = f_{\theta^*}(x)$  (shape  $1 \times H \times S$ )
- 4: Extract:  $NH_4 = \hat{y}[:, :, o]$ ,  $SS = \hat{y}[:, :, 4]$ ,  $COD = 1.42 \times SS$
- 5: Compute cost:
$$J = \mathbb{E}[NH_4^2] + 0.5 \cdot \mathbb{E}[COD^2] + 0.01 \cdot u_o^2$$
- 6: **return**  $J$

By leveraging accurate predictions from the Transformer model, the MPC controller can optimize control inputs while satisfying operational constraints.

#### Algorithm 4: Transformer-Based MPC Closed-Loop Control

**Require:**

Trained model  $f_{\theta^*}$ , initial state window  $x_o \in \mathbb{R}^{1 \times T \times F}$ , MPC steps  $K = 50$ , horizon  $H = 10$

Control bounds  $u \in [0, 5]$ , control index  $u_{idx} = 12$

- 1: Initialise:  $x \leftarrow x_o$ ; history buffers  $\mathcal{U}, \mathcal{N}_{NH_4}, \mathcal{S}_{SS}, \mathcal{S}_{O_2} \leftarrow []$
- 2: **for**  $k = 1, \dots, K$  **do**
- 3:   Sample initial guess  $U_o \sim \text{Uniform}([0.5, 4.0]^H)$
- 4:   Solve:  $U^* = \arg \min_{U \in \mathcal{U}} J(U; \cdot, f_{\theta^*}, x)$  via Powell, using **Algorithm 3** as objective
- 5:   Apply first control:  $u_k = U_o^*$ ; append to  $\mathcal{U}$
- 6:   Set  $x[:, -1, u_{idx}] \leftarrow u_k$
- 7:   Predict next states:  $\hat{y}_k = f_{\theta^*}(x[:, :, :])$  (first horizon step)
- 8:   Record: append  $NH_4, SS, O_2$  from  $\hat{y}_k$  to respective history buffers
- 9:   Shift window:  $x \leftarrow \text{roll}(x, -1, \text{axis}=\text{dim}-1)$ ; set  $x[:, -1, :S] \leftarrow \hat{y}_k$
- 10: **end for**
- 11: **return** histories  $\mathcal{U}, \mathcal{N}_{NH_4}, \mathcal{S}_{SS}, \mathcal{S}_{O_2}$

## VI. RESULTS AND DISCUSSION

### A. Experimental Setup

The proposed model is evaluated using a large-scale WWTP dataset. The dataset is divided into training and testing subsets, where the training set is used for model learning and the testing set consists of unseen samples for performance evaluation.

### B. Performance Metrics

To quantitatively assess prediction accuracy, the following metrics are used.

**Algorithm 2: Model Evaluation on Held-Out Test Set**

**Require:**

Trained model  $f_{\theta^*}$ , test set  $\mathcal{D}_{test}$ , state map  $\{NH_4:0, NO_3:1, O_2:2, PO_4:3, SS:4\}$

- 1: Set model to evaluation mode; compute  $\hat{Y}_{test} = f_{\theta^*}(X_{test})$  with no gradient computation
- 2: Flatten predictions and targets to shape  $(N \cdot H, S)$
- 3: Compute overall metrics:  $MSE, RMSE = \sqrt{MSE}, MAE, R^2$
- 4: Derive effluent proxies at timestep  $t = \sigma$ :  
 $COD = 1.42 \times SS, BOD = 0.65 \times COD$
- 5: **return** MSE, RMSE, MAE,  $R^2$ , COD and BOD predictions

The mean squared error (MSE) is defined as

$$MSE = \frac{1}{N} \sum_{k=1}^N \|Y_k - \hat{Y}_k\|^2 \quad (13)$$

The root mean squared error (RMSE) is given by

$$RMSE = \sqrt{MSE} \quad (14)$$

The mean absolute error (MAE) is defined as

$$MAE = \frac{1}{N} \sum_{k=1}^N |Y_k - \hat{Y}_k| \quad (15)$$

The coefficient of determination is expressed as

$$R^2 = 1 - \frac{\sum (Y_k - \hat{Y}_k)^2}{\sum (Y_k - \bar{Y})^2} \quad (16)$$

### C. Prediction Results

The evaluation results demonstrate that the proposed Transformer model achieves low prediction errors and high  $R^2$  values on unseen test data, indicating strong generalization capability.

The predicted trajectories closely follow the actual system behavior, particularly for key variables such as ammonium ( $NH_4$ )

concentration. The model effectively captures both steady-state behavior and transient dynamics.

### D. Analysis of Temporal Dynamics

From the time-series plots, it is observed that:

1. The predicted outputs exhibit minimal deviation from the ground truth.
2. The model successfully captures rapid fluctuations in system states.
3. No significant prediction lag is observed across the prediction horizon.

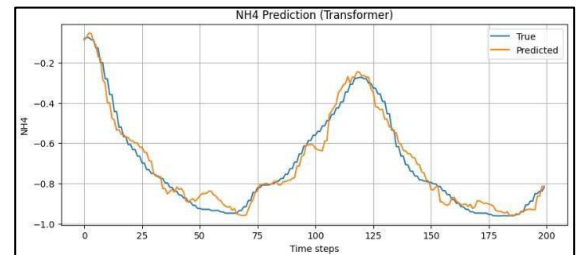
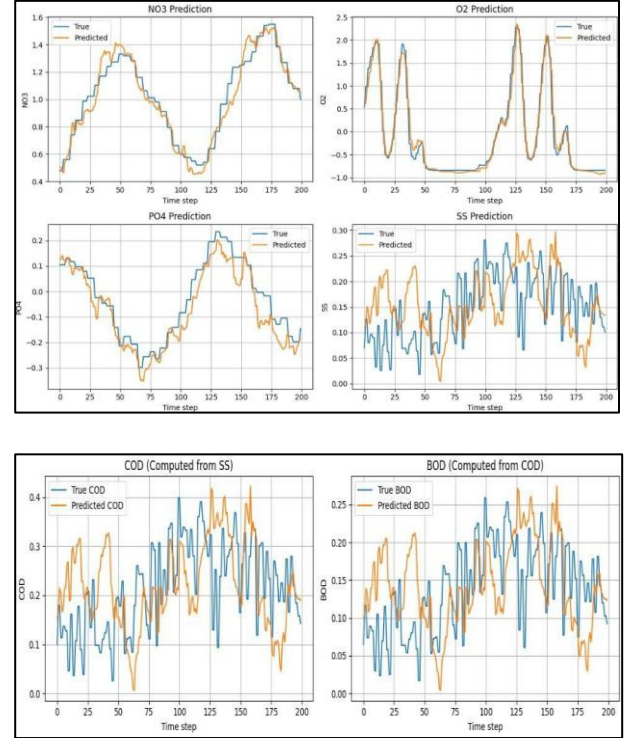


Fig 2 Transformer Results for NO3, O2, PO4, NH4, SS, COD, BOD



These results confirm that the Transformer architecture is capable of learning complex temporal dependencies in WWTP processes.

## E. Discussion on Control Performance

The accurate multi-step predictions provided by the proposed model significantly enhance the performance of the MPC controller.

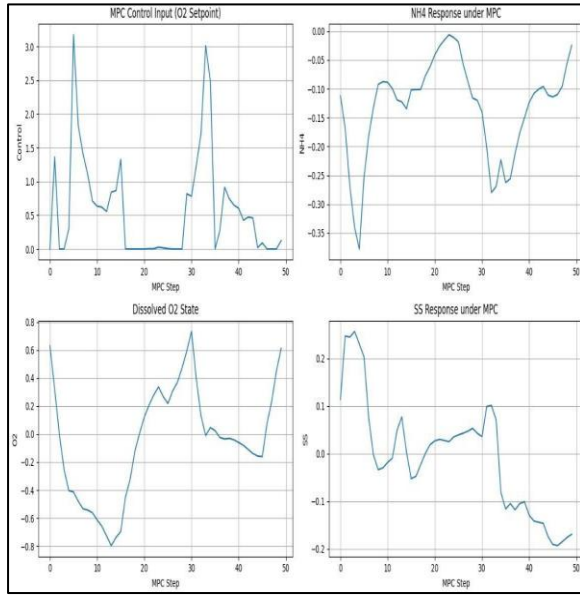


Fig 3 MPC Results on Controlled State

Specifically:

- The tracking error  $z_k$  is reduced due to improved prediction accuracy.
- Control actions are optimized more effectively over the prediction horizon.
- The system shows improved robustness against disturbances.

## F. Comparative Discussion

Compared with conventional approaches, the proposed method offers several advantages:

1. Traditional PID controllers lack adaptability to nonlinear dynamics.
2. First-principles models require extensive domain knowledge and calibration.
3. Standard neural networks fail to capture long-term temporal dependencies.

In contrast, the Transformer-based model provides a unified framework for learning complex multivariable relationships directly from data.

## G. Limitations and Future Work

Despite its effectiveness, the proposed approach has certain limitations:

1. High computational cost during training.
2. Dependence on large-scale datasets.
3. Limited interpretability compared to physical models.

Future work will focus on integrating physics-based constraints and reinforcement learning strategies to further enhance control performance.



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